CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 3: **Pair of Linear Equations in Two Variables** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The pair $5x + ky = 15$, $10x + 10y = 30$ has infinitely many solutions when $k =$ 1

(a) 5 **(b)** 15 **(c)** 6 **(d)** 10

2. For the pair $kx + 5y = 1$ and $6x + 2y = 7$ to have a unique solution, k must NOT equal: 1

(a) $\frac{12}{5}$ **(b)** 11 **(c)** $\frac{6}{7}$ **(d)** 15

3. If $6x + 5y = 52$ and $5x + 6y = 49$, then $x + y =$ 1

(a) $\frac{101}{2}$ **(b)** 101 **(c)** $\frac{101}{11}$ **(d)** 3

4. If $x + y = 15$ and $x - y = 7$, then $x^2 - y^2 =$ 1

(a) 105 **(b)** 22 **(c)** 176 **(d)** 274

5. The value of x in $8x + 3y = 30$, $3x + 5y = 19$ is: 1

(a) 3 **(b)** 2 **(c)** -3 **(d)** 5

6. The value of y in $15x + 4y = 66$, $7x + 6y = 68$ is: 1

(a) -9 **(b)** 2 **(c)** 11 **(d)** 9

7. The pair $6x + ky = 18$, $12x + 12y = 36$ has infinitely many solutions when $k =$ 1

(a) 6 **(b)** 18 **(c)** 7 **(d)** 12

8. For the pair $kx + 2y = 9$ and $5x + 4y = 3$ to have a unique solution, k must NOT equal: 1

(a) 10 **(b)** $\frac{5}{2}$ **(c)** 7 **(d)** 15

9. If $7x + 3y = 60$ and $3x + 7y = 40$, then $x + y =$ 1

(a) 100 **(b)** 50 **(c)** 10 **(d)** 5

10. If $x + y = 12$ and $x - y = 6$, then $x^2 - y^2 =$ 1

(a) 180 **(b)** 108 **(c)** 72 **(d)** 18

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find k so that the pair $5x + ky = 15$ and $10x + 10y = 30$ has infinitely many solutions. 2

12. Solve by the cross-multiplication method: $9x + 2y = 50$ and $5x + 3y = 41$. 2

13. For what value of k does the pair $kx + 5y = 1$, $6x + 2y = 7$ have a unique solution? 2

14. Find k so that the pair $7x + ky = 21$ and $14x + 14y = 42$ has infinitely many solutions. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve by elimination: $8x + 3y = 30$ and $3x + 5y = 19$. 3

16. The sum of the digits of a two-digit number is 12. If the digits are interchanged, the number becomes 36 more. Find the number. 3

17. Find the value of k for which the pair $kx + 5y = k - 5$ and $20x + ky = k$ has infinitely many solutions. 3

OR

Solve by elimination: $4x - 7y = -10$ and $5x + 6y = 76$. 3

18. The sum of the digits of a two-digit number is 12. If the digits are interchanged, the number becomes 54 more. Find the number. 3

OR

Find the value of k for which the pair $kx + 6y = k - 6$ and $24x + ky = k$ has infinitely many solutions. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The taxi charges consist of a fixed charge plus a charge per km. For 10 km the fare is ₹130 and for 25 km it is ₹250. Find the fixed charge, the charge per km and the fare for 40 km. 5

OR

If the length of a rectangle is reduced by 6 units and its breadth is increased by 3 units, its area decreases by 42 square units. If the length is increased by 4 units and the breadth by 2 units, the area increases by 120 square units. Find the length and breadth. 5

20. The taxi charges consist of a fixed charge plus a charge per km. For 6 km the fare is ₹125 and for 14 km it is ₹245. Find the fixed charge, the charge per km and the fare for 22 km. 5

OR

If the length of a rectangle is reduced by 2 units and its breadth is increased by 3 units, its area increases by 18 square units. If the length is increased by 3 units and the breadth by 1 units, the area increases by 44 square units. Find the length and breadth. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND